Tzuyang (Frank) Lin

LinkedIn | GitHub | zzyang0.github.io | franklin532232@gmail.com | Shanghai, CN

Data Analyst | SQL | Python | Tableau | ETL | Business Intelligence | Data Visualization

EDUCATION

University of California, Berkeley — **Berkeley, California**

Major: Master of Analytics (GPA: 3.88/4.00) — Graduation Date: Aug 2025

Coursework: Analysis and Design of Databases, Machine Learning, Optimization Analytics (Gurobi), Data Modeling and Statistics

University of Washington, Seattle — **Seattle, Washington**

Major: B.A. in Economics / Minor: Informatics (GPA: 3.85/4.00) — Graduation Date: Mar 2024

Coursework: Data Science for Strategic Pricing, Computational Finance & Econometrics, Portfolio Optimization, Information Visualization, Database Management Systems

WORK EXPERIENCE

OnPoint Insights — **Remote, US**

Data Analyst — Oct 2025 – Jan 2026

- Cleaned and processed Mass Save energy efficiency datasets using Python, thoroughly comparing planned versus actual Benefit-Cost Ratios (BCR) to validate ROI compliance and empower project performance evaluations.
- Developed frameworks for heat pump installation trends and user segmentation, identifying 40%+ year-over-year growth dynamics and coverage gaps within low-income demographics.
- Built an end-to-end conversion funnel for residential energy retrofits, quantifying a 15% target gap and pinpointing core drop-off bottlenecks to deliver targeted optimization recommendations.

Takachar — **Petaluma, CA**

Data Dashboard & Web App Developer Intern — Jun 2025 – Aug 2025

- Engineered a web-based data collection platform using Streamlit and Python integrating camera capture, geolocation, and QR scanning, reducing stakeholder submission time by 30% and improving carbon credit reporting procedures across pilot regions.
- Built an automated ETL pipeline ingesting over 40K environmental records into a Tableau dashboard with line charts and scatter plots, improving operational efficiency and enabling leadership to monitor sequestration trends and compliance KPIs weekly.
- Deployed a Python data migration pipeline connecting Kobo Toolbox, Azure SQL, and SharePoint, eliminating manual file transfers and improving data consistency for carbon verification workflows to support scalable field site onboarding without additional engineering resources.

Big Data Investment Development Co. — **Ningbo, China**

Data Analyst Intern — Jul 2023 – Sep 2023

- Processed over 100K provincial and municipal investment records using MaxCompute and Python scripts to impute missing values and annotate datasets in Excel, enabling management teams to monitor data quality and maintain clean reporting pipelines.
- Optimized query performance using multi-table SQL joins, window functions, CASE statements, and encryption functions, achieving 7% efficiency improvement that reduced data retrieval time across business units and accelerated ad-hoc analysis requests from senior leadership.
- Architected ETL pipelines transforming Operational Data Store to Data Warehouse architecture using MaxCompute, streamlining data accessibility and establishing scalable infrastructure for future integration initiatives.

PROJECT EXPERIENCE

R-based Portfolio Return and Risk Assessments — **Seattle, WA**

Business & Data Analyst — Mar 2023 – Jun 2023

- Collected 60 months of historical price data for five Vanguard ETFs using R with quantmod, calculating continuously compounded returns and conducting normality tests to validate time-series modeling assumptions for risk-adjusted portfolio recommendations to wealth management clients.
- Computed Value-at-Risk at 1% and 5% levels using bootstrap and empirical methods, quantifying losses from $2,100 to $5,052 to inform allocation decisions and compare risk-return profiles across investment strategies.
- Built efficient frontier models in R comparing portfolios with and without short-selling, reducing volatility while achieving a 13.6% annual return and delivering visualizations demonstrating diversification benefits for investment recommendations.

TECHNICAL SKILLS

Programming Languages: Python, R, JavaScript, CSS, SQL, HTML, Gurobi

Framework & Libraries: Bootstrap, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Tableau, Git

Databases & Cloud Platforms: MySQL, Azure SQL, MaxCompute, SharePoint

Technical Skills: ETL Pipelines, Data Warehousing, Statistical Modeling, A/B Testing, Time Series Analysis, Predictive Modeling